

The sublimation kinetic coefficient was inflated by an inapplicable thin-interface correction

What was wrong, how it was found, what was changed, and why it works

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Summary. The solver did not deliver the attachment kinetics it was given. τ_{sub} , the phase-field relaxation time, carried two Karma thin-interface counter-terms intended to cancel a spurious contribution generated at finite interface width. For this model that contribution is zero, because vapor diffuses only in the pore, so nothing cancelled and the realized kinetic coefficient came out as $\beta_{\text{sub0}} + \Delta$ instead of β_{sub0} . At -20°C with $\epsilon = 8.58 \times 10^{-7}$ m the excess Δ was 22 %, and because Δ is proportional to ϵ the realized kinetics depended on the mesh. The defect was found by fitting the Gibbs–Thomson relation directly to nine wedge-channel runs, which recovered the capillary length exactly but put the kinetic coefficient 22 % high; it was confirmed by a sweep in β_{sub0} over a range of 40, which reproduced the predicted shortfall from 0.32 to 0.95. The fix is to remove the two terms (`-thin_iface_corr`, default 0). After it, five runs spanning the same range recover the requested kinetics to within 1 %, and a mirror-image control reproduces both menisci to 0.1 %. One check remains outstanding: that the answer no longer moves when ϵ is refined.

1. What the solver is supposed to do

The model advances an ice phase field ϕ (1 in ice, 0 in vapor) coupled to a vapor density ρ_v . At the ice–vapor interface it is intended to satisfy the Gibbs–Thomson condition

$$\sigma = \beta_{\text{sub0}} v_n + d_{0,\text{sub0}} \chi, \quad \sigma = \frac{\rho_v - \rho_{vs}(T)}{\rho_{vs}(T)}, \quad (1)$$

in which σ is the supersaturation, $\rho_{vs}(T)$ the saturation vapor density, v_n the interface normal velocity (positive for growth), and χ the sum of the principal curvatures (positive where the ice is convex into the vapor). The two coefficients are user inputs: `-beta_sub0`, the attachment kinetic coefficient, which is set from the condensation coefficient α_c through the Hertz–Knudsen relation $\beta_{\text{sub0}} \propto 1/\alpha_c$; and `-d0_sub0`, the capillary length, set by the ice surface energy. Equation (1) is the target physics and is not in question anywhere in this note.

The phase-field equation that is supposed to produce Eq. (1) is

$$\tau_{\text{sub}} \partial_t \phi = \epsilon^2 \nabla^2 \phi - f'(\phi) + \lambda_{\text{sub}} g(\phi) \frac{\rho_{vs}(T)}{\rho_{\text{ice}}} \sigma, \quad (2)$$

where ϵ is the interface parameter, f the double well, $g(\phi) = \phi^2(1 - \phi)^2$ the coupling function, and λ_{sub} and τ_{sub} are chosen so that the sharp-interface limit of Eq. (2) reproduces Eq. (1) with the requested $d_{0,\text{sub0}}$ and β_{sub0} .

2. What was wrong

2.1 The counter-terms

Taking $\epsilon \rightarrow 0$ in Eq. (2) gives Eq. (1) with

$$d_{0,\text{realized}} = d_{0,\text{sub0}}, \quad \beta_{\text{realized}} = \frac{\tau_{\text{sub}} d_{0,\text{sub0}}}{\epsilon^2} - C_{\text{corr}}. \quad (3)$$

C_{corr} is a *spurious* $O(\epsilon)$ term: at any finite interface width σ varies across the interfacial region where the phase change happens, and that variation feeds back through the $\lambda_{\text{sub}}g(\phi)\sigma$ coupling to add an unwanted kinetic contribution. The standard remedy, due to Karma & Rappel (Phys. Rev. E **57**, 4323, 1998), is to inflate τ_{sub} by exactly the expected C_{corr} so that the two cancel in Eq. (3). The solver did this:

$$\tau_{\text{sub}} = \epsilon \lambda_{\text{sub}} \left(\underbrace{\frac{\beta_{\text{sub}}}{a_1}}_{\text{the kinetics requested}} + \underbrace{\frac{a_2 \epsilon}{\text{diff_sub}}}_{\text{thermal counter-term}} + \underbrace{\frac{a_2 \epsilon}{\text{dif_vap}}}_{\text{vapor counter-term}} \right), \quad (4)$$

with $a_1 = 5$ and $a_2 = 0.1581$ the constants for this model's double well, `diff_sub` the mean thermal diffusivity and `dif_vap` the vapor diffusivity. The bracket is written in the $\rho_{\text{vs}}/\rho_{\text{ice}}$ -scaled variables the solver carries internally.

2.2 The consequence

If C_{corr} is not in fact produced, nothing cancels, the inflation survives, and the realized coefficient is

$$\beta_{\text{realized}} = \beta_{\text{sub0}} + \Delta, \quad \Delta = a_1 a_2 \epsilon \left(\frac{1}{\text{diff_sub}} + \frac{1}{\text{dif_vap}} \right) \frac{\rho_{\text{ice}}}{\rho_{\text{vs}}(T)}. \quad (5)$$

Δ is the central quantity of this note: the excess kinetic coefficient, in the same units as β_{sub0} (s m^{-1}). At -20°C with $\epsilon = 8.584 \times 10^{-7} \text{ m}$ it is $1.282 \times 10^5 \text{ s m}^{-1}$, against a requested $\beta_{\text{sub0}} = 5.9216 \times 10^5$: an excess of 22 %, of which three quarters comes from the thermal term.

3. How it was found

3.1 Measurement

The wedge geometry places an apex-centered ice annulus in a tapering pore, with the virtual apex on the channel axis. On the centerline the interface normal is therefore exactly $\pm \hat{x}$ and v_n reduces to a one-dimensional displacement rate, which makes it measurable without reconstructing normals or differentiating contours. `postprocess/wedge_gt_velocity.py` measures σ , χ and v_n on both menisci of each run.

3.2 Fitting both coefficients

Rather than assume Eq. (1) and report a ratio, both coefficients were fitted. Each snapshot gives a triple (σ, v_n, χ) , so $\sigma = \beta v_n + d_0 \chi$ is a two-column least-squares problem. It is identifiable because the two menisci carry curvature of opposite sign, the inner being concave and the outer convex, while a 3×3 grid of vapor boundary conditions varies v_n at nearly fixed χ (condition number 2.58). Over nine runs, $n = 3492$:

	fitted	relative to requested
d_0	$1.0172 \times 10^{-9} \text{ m}$	1.0006
β	$7.2324 \times 10^5 \text{ s m}^{-1}$	1.2214

with $R^2 = 0.999997$ and an rms residual of 0.14 % of $|\sigma|$. Two things follow. The relation itself holds: a different functional form would not fit this well. And the capillary length is delivered exactly, so the discrepancy is confined to β . Since β and d_0 are separately identified, they are not trading off; d_0 came out correct both before and after the fix, while β moved by 22 %.

The fitted β agrees with Eq. (5) to 0.4 %, which identifies the counter-terms as the source.

3.3 Confirming it

Because Δ is additive and does not scale with β_{sub0} , the predicted shortfall $\beta_{\text{sub0}}/(\beta_{\text{sub0}} + \Delta)$ is a strong function of β_{sub0} . Four runs differing only in `-beta_sub0`, each 3406 steps (batch 2026-08-25__18.14.15_beta_ladder):

β_{sub0} relative	predicted ratio	measured inner	measured outer	$\beta_{\text{fit}}/(\beta_{\text{sub0}} + \Delta)$
0.10×	0.3160	0.3239	0.3082	1.0004
0.25×	0.5359	0.5444	0.5254	1.0015
1.00×	0.8220	0.8253	0.8126	1.0017
4.00×	0.9487	0.9471	0.9429	1.0000

The ratio moves by a factor of 2.9 and follows the prediction to 2.3 %. A constant 1.00, meaning the counter-terms work as intended, and a constant 0.82, meaning a fixed factor of some other origin, are both excluded by every run. Fitting β with d_0 held fixed gives $\beta_{\text{fit}}/(\beta_{\text{sub0}} + \Delta) = 1.0009 \pm 0.0007$, so $C_{\text{corr}} = 0$ to better than one part in 10^3 .

4. Why the correction is absent

The Karma & Rappel counter-term is derived for a *symmetric* model, meaning the diffusing quantity has the same diffusivity in both phases. Our vapor equation instead carries $D_v \phi_a$, where $\phi_a = 1 - \phi$: vapor diffuses in the pore and not in the ice. This is the *one-sided* case, and it behaves differently.

Write ζ for signed normal distance from the interface and $\eta = \zeta/\epsilon$ for the stretched inner coordinate. Expanding $\phi = \phi_0(\eta) + \epsilon \phi_1 + O(\epsilon^2)$, where the subscript is the order in ϵ and not a phase label, the leading term $\phi_0 = \frac{1}{2}(1 + \tanh(\eta/2))$ is the planar equilibrium profile. Splitting $\sigma = \sigma_{\text{int}} + \delta\sigma(\eta)$, with σ_{int} the outer solution extrapolated to the interface, the counter-term is proportional to

$$I = \int g(\phi_0) \partial_\eta \phi_0 \delta\sigma(\eta) d\eta. \quad (6)$$

Neglecting storage, which is smaller than diffusion by 2.4×10^{-11} here, the inner vapor balance integrates once to

$$D_v \phi_a \partial_\eta \rho_v = \epsilon v_n \rho_{\text{ice}} \phi_a \implies \partial_\eta \rho_v = \frac{\epsilon v_n \rho_{\text{ice}}}{D_v} \quad \text{wherever } \phi_a \neq 0, \quad (7)$$

the constant of integration being fixed by the absence of flux into the ice. The factor ϕ_a cancels, and the surviving constant is exactly the outer Stefan gradient. The inner solution is therefore the outer solution continued through the interface, $\delta\sigma \equiv 0$, and I vanishes for any weight. With a symmetric D the ϕ_a does not cancel, $\delta\sigma$ is non-zero and even in η , and I survives; this is the case Karma & Rappel treat.

Two remarks. First, the thermal counter-term fails for an unrelated reason: `a2*eps/diff_sub` treats latent heat as driving σ at unit strength, whereas temperature reaches σ only through $\rho_{vs}(T)$ and so carries a Clausius–Clapeyron factor. The correct coefficient, $(d\rho_{vs}/dT)L_{\text{sub}}/k$, is smaller by a factor of 11 to 1300. Second, the argument above bounds only the terms proportional to v_n . A one-sided model without an anti-trapping current can also produce a spurious surface diffusion, which is a different operator and is not observable in this measurement; see §8.

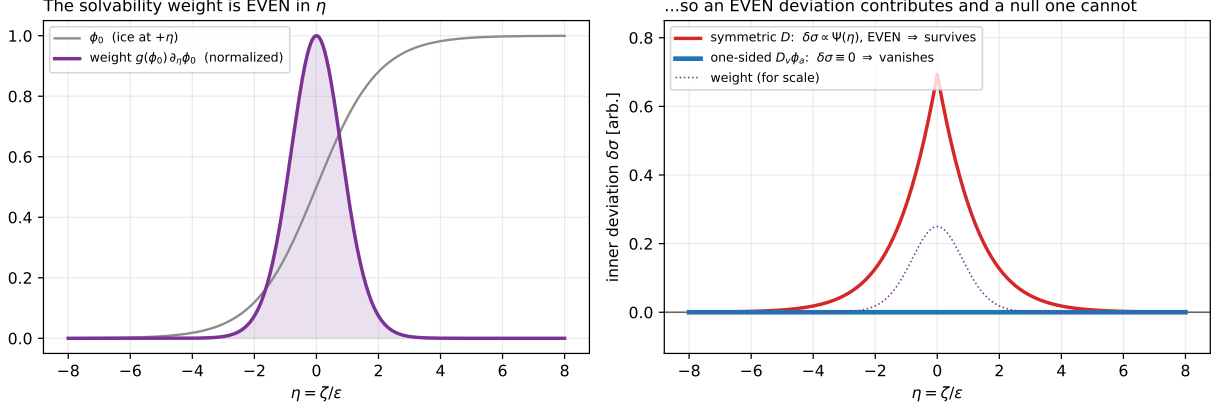


Figure 1: Left: the weight $g(\phi_0) \partial_\eta \phi_0$ in Eq. (6) is even in η . Right: with symmetric D the inner deviation $\delta\sigma$ is even and non-zero and the counter-term survives; with the one-sided $D_v \phi_a$ it vanishes identically, so no counter-term can arise.

4.1 The resulting definitions of β and τ_{sub}

With $C_{\text{corr}} = 0$, Eq. (3) reduces to

$$d_{0,\text{realized}} = d_{0,\text{sub}0}, \quad \beta_{\text{realized}} = \frac{\tau_{\text{sub}} d_{0,\text{sub}0}}{\epsilon^2}, \quad (8)$$

so the relaxation time that realizes a requested $\beta_{\text{sub}0}$ follows by inversion:

$$\boxed{\tau_{\text{sub}} = \frac{\epsilon^2 \beta_{\text{sub}0}}{d_{0,\text{sub}0}}} \quad \text{equivalently} \quad \tau_{\text{sub}} = \epsilon \lambda_{\text{sub}} \frac{\beta_{\text{sub}}}{a_1}, \quad (9)$$

the second form being the one the code evaluates. Two properties are worth noting. The $\rho_{vs}/\rho_{\text{ice}}$ scaling cancels between β_{sub} and $d_{0,\text{sub}}$, so τ_{sub} depends only on the unscaled inputs and enters temperature only through them. And $\tau_{\text{sub}} \propto \epsilon^2$: since the stable time step follows τ_{sub} , halving ϵ quarters `-dtmax`, which is what the convergence arm of §8 must do.

For contrast, a symmetric diffusivity would require

$$\tau_{\text{sub}} = \frac{\epsilon^2 (\beta_{\text{sub}0} + \Delta)}{d_{0,\text{sub}0}}, \quad (10)$$

and Eq. (10) is what was coded. The difference between the two is exactly the counter-term: at -20°C with $\epsilon = 8.584 \times 10^{-7}$ m, Eq. (9) gives $\tau_{\text{sub}} = 429.2$ s and Eq. (10) gives 522.1 s.

5. The change

Both counter-terms are now behind `-thin_iface_corr`, which defaults to 0, so that the solver evaluates Eq. (9) rather than Eq. (10). The flag is present in the scalar path of both solvers and in the pointwise path `SubKinetics`, with the analytic Jacobian derivatives gated alongside them. Setting the flag to 1 restores the terms in their historical form for comparison with earlier runs; it is not a corrected form, and since the physically correct counter-term for this model is approximately zero, no corrected form is provided. Each run now reports the realized coefficient in its header.

Removing the terms is equivalent to supplying $\beta_{\text{sub}0} - \Delta$ in their presence, but is preferable: it is applied once rather than in every run configuration, cannot be invalidated by a later change of ϵ or T , and remains defined when $\Delta > \beta_{\text{sub}0}$, where no pre-compensation exists. Rescaling $\beta_{\text{sub}0}$ by the observed ratio would not work at all, since that ratio is $\beta_{\text{sub}0}/(\beta_{\text{sub}0} + \Delta)$ and varies from 0.32 to 0.95 over the range tested.

6. That the fix works

Five runs, batch 2026-09-09__09.22.56_tau_fix, with no solver failures:

run	before	after		σ sensitivity
		inner	outer	
0.10×	0.3160	1.0863	0.9539	9.5 % / 3.3 %
0.25×	0.5359	1.0330	0.9720	3.6 % / 1.4 %
1.00×	0.8220	1.0056	0.9877	0.9 % / 0.3 %
4.00×	0.9487	0.9985	0.9936	0.2 % / 0.1 %

Fitting β with d_0 held fixed gives $\beta_{\text{fit}}/\beta_{\text{sub0}}$ of 0.9916, 1.0012, 1.0017 and 1.0003. The residual scatter tracks the last column run for run and is therefore measurement rather than model. It is larger at small β_{sub0} for a physical reason: with the correct, smaller β the interface sits nearer local equilibrium, so the residual $\sigma - d_{0,\text{sub0}}\chi$ shrinks and a fixed absolute error in σ becomes a larger fraction of it. Precision is now best at large β_{sub0} , the reverse of before.

Mirror control. A fifth run reflects the channel about $x = L_x/2$, putting the wide end at $x = 0$ and exchanging the two vapor reservoirs with it, so that the concave meniscus moves from the left of the ice to the right. The two menisci exchange values exactly:

	inner (concave)	outer (convex)
unmirrored	1.0056	0.9877
mirrored	1.0056	0.9869

with $d_{0,\text{sub0}}\chi$ agreeing to 0.07 %. The residual 1 % difference between the two menisci therefore follows the geometry rather than the side, and is a property of curvature rather than an artifact of the mesh or the measurement.

7. How this changes the model’s behavior

The change alters the relationship between what is requested and what is realized. Every statement below concerns the *old* behavior; after the change the realized coefficient is β_{sub0} and none of these dependencies remain.

Attachment coefficient α_c . Since $\beta_{\text{sub0}} \propto 1/\alpha_c$ while Δ does not depend on α_c at all, the realized α_c was always lower than requested, and increasingly so as α_c was raised. Worse, $\beta_{\text{realized}} \geq \Delta$ placed a ceiling on the attainable α_c that no input could exceed. At $\epsilon = 8.584 \times 10^{-7}$ m that ceiling was 0.062 at -20°C and 0.0006 at -60°C . Tuning α_c against experiment was therefore adjusting a knob that did not do what it said, and hardest to move where the physics most wanted it moved.

Interface parameter ϵ . $\Delta \propto \epsilon$, so the realized kinetics depended on the mesh. This is the most serious consequence: a converged sharp-interface limit cannot depend on ϵ , so no mesh refinement could ever have converged. Halving ϵ would have moved the ratio from 0.822 to 0.902.

Diffusivities. $\Delta \propto 1/\text{diff_sub} + 1/\text{dif_vap}$, so raising either reduced the error. This matters for the Molaro study, where the $D_v \times 10$ arm was chosen to suppress transport resistance: because the thermal term dominates Δ and does not involve D_v , that arm concentrated the error instead of diluting it, and its rate loss was 10.7 % against 4.2 % for the nominal- D_v arm.

Temperature. $\Delta \propto 1/\rho_{vs}(T)$, so the error grew rapidly as temperature fell: 22 % at -20°C , a factor of 1.7 at -40°C , and a factor of 20.7 at -60°C .

What is unaffected. Scaling exponents and dimensionless ratios, since they do not depend on the absolute rate. Quantities that do depend on it, such as times to a given state or absolute growth rates, are displaced by $\beta_{\text{sub0}}/(\beta_{\text{sub0}} + \Delta)$ for the (ϵ, T, α_c) of the run in question, and any earlier conclusion resting on one should be re-checked. `preprocess/comp_eps.py` sizes ϵ using β_{sub0} ; with the counter-terms gone that relation is no longer implicit.

8. Outstanding

ϵ -convergence, staged and not yet run. All of the above appeals to our own analysis at some point. The one check that does not is refinement: a converged sharp-interface limit cannot depend on ϵ . `tests_eps_convergence.txt` halves ϵ to 4.292×10^{-7} m (989×659 , four times the cells) and predicts the same ratio as the existing ϵ run, 1.0056 and 0.9877. The old τ_{sub} would instead have moved it to 0.9023. Until this runs, the claim that the fix gives the correct sharp-interface limit rests on the derivation of §4 rather than on measurement.

Spurious surface diffusion. A one-sided model without an anti-trapping current can generate an effective surface diffusion at $O(\epsilon)$. It would appear nowhere in the source, and nothing here bounds it, because it is a tangential operator invisible to a centerline v_n measurement. The surface excess of the diffusivity weight, $\int [(1 - \phi_0) - \Theta(-\eta)] d\eta$, vanishes exactly by the same antisymmetry as I , which suggests it is suppressed, but that is an indication rather than a calculation. Settling it means either the one-sided asymptotics for this model, mapping onto Echebarria, Folch, Karma & Plapp (Phys. Rev. E **70**, 061604, 2004), or adding the anti-trapping current of Karma (Phys. Rev. Lett. **87**, 115701, 2001). Neither is urgent, since the effects such a current cancels scale as $\epsilon v_n / D_v$, here 2.7×10^{-14} . It matters specifically because the model contains no surface-diffusion term by construction, while the Molaro analysis separately invokes physical surface diffusion; the two must not be conflated.